Fluid Lines and Fittings – Torquing

1. General

Unless otherwise detailed in individual maintenance practices this standard specifies procedures and requirements for torquing fluid line tapered thread fittings, AN818 and NAS591 fluid line coupling nuts.

2. Materials and Equipment

A. Materials:

- Castrol anti-seize compound with zinc oxide base
- SAE AMS 2518
- ROYCO 1MS MIL–PRF–83483
- SAE AMS–G–4343 (e.g., ROYCO 43)
- VV–P–236 (e.g., PENRECO Amber Petrolatum)
- Titeseal #2 or #3
- Monsanto MCS 352
- Allube HI–LO MS #1

B. Equipment:

- Torque wrenches, sockets and adapters as required

3. Lubricate Threads

A. Refer to Table 1 — Fluid Lines and Fittings Thread Lubricants for required lubricant.

B. Apply lubricant sparingly to male threads only (to prevent clogging of system or contamination of fluid).

C. On straight threads, apply lubricant to all but the first two threads.

D. On tapered threads (except oxygen and pitot-static systems), apply lubricant to the first three threads and ensure that no lubricant enters the bore of the fitting.

WARNING DO NOT LUBRICATE OXYGEN SYSTEM FITTINGS. OIL OR GREASE EXPOSED TO OXYGEN UNDER PRESSURE COULD CAUSE AN EXPLOSION RESULTING IN SERIOUS INJURY TO PERSONNEL AND/OR DAMAGE TO EQUIPMENT.

E. On oxygen and pitot-static system tapered threads, wrap the thread circumference once with teflon tape, cut tape length to obtain a minimum overlap, and work into threads with fingers.

F. When the system fluid is specified in Table 1 — Fluid Lines and Fittings Thread Lubricants, as the lubricant, the fluid may be applied generously.

Table 1 Fluid Lines and Fittings Thread Lubricants

Line Content	Type of Thread	Thread Lubricant
Fuel or Oil	Straight Thread	Castrol anti-seize compound or PENRECO amber petrolatum
	Tapered Thread	Titeseal #2 or #3
Hydraulic fluid (petroleum base)	Straight Thread	PENRECO amber petrolatum or Aeroshell fluid #4 or Univis J43
	Tapered Thread	PENRECO amber petrolatum

Table 1 Fluid Lines and Fittings Thread Lubricants (continued)

Line Content	Type of Thread	Thread Lubricant
Hydraulic fluid (synthetic phosphate ester)	Straight Thread	Monsanto MCS 352 or Allube HI–LO MS #1 or Hydraulic fluid (synthetic phosphate ester) or Phosphocent 246
	Tapered Thread	MCS 352 (Monsanto) or HI–LO MS #1 (allube) or Phosphocent 246
Water/Methanol	All	PENRECO amber petrolatum
Oxygen	Straight Thread	None
	Tapered Thread	Teflon tape
Pitot – Static	Straight Thread	AMS–G–4343
	Tapered Thread	Teflon tape
Compressed air (not oxygen)	All	AMS–G–4343
Hot gases (up to 800°F)	All	MIL–PRF–83483
Other fluids (up to 300°F)	All	Castrol anti-seize

4. Torquing

A. Torque Oxygen and Pitot System Tapered Thread Fittings

1) Screw fitting in not less than two and one half turns then torque between 22 and 30 pounds–inches.

2) Replace fittings that leak when torqued to 30 pound–inches.

3) Remove and discard any fitting torqued to more than 30 pounds–inches. Do not loosen and retorque.

B. Torque Fluid Lines and Fittings

1) Refer to Table 2 — Torque Coupling Nuts on Flared Tubes and Fittings for Torque Values.

2) Correct size and proper fitting wrenches must be used.

3) Torque fittings to a minimum value indicated in Table 2 — Torque Coupling Nuts on Flared Tubes and Fittings.

4) Torque fittings to maximum value indicated in Table 2 — Torque Coupling Nuts on Flared Tubes and Fittings only to adjust a position type fitting or to correct a leaking connection.

5) Remove and discard fitting or line on connections that continue to leak after being torqued to maximum value indicated in Table 2 — Torque Coupling Nuts on Flared Tubes and Fittings.

6) Remove and discard overtorqued fittings. Do not loosen and retorque.

7) When torquing the nut on a union, restrain union with a wrench while torquing nut.

8) Ensure that flexible hoses and rigid lines are not twisted or under preload during torquing.

Table 2 Torque Coupling Nuts on Flared Tubes and Fittings

SIZE (Note 2)	TORQUE VALUE (in–lbs) (Note 1)		
	STRAIGHT THREADS ON ALUMINUM ALLOY TUBE OR FITTING	**STRAIGHT THREADS ON STEEL ALLOY, CRES OR TITANIUM TUBE OR FITTINGS**	**TAPERED PIPE THREADS (except oxygen & pitot systems)**
1/8”	20 – 30	75 – 85	40 – 300
3/16”	25 – 35 (15 – 35 for 5052–0 alu-minum tubing with a wall thick-ness of 0.022”)	95 – 105	—
1/4”	50 – 65 (20 – 65 for 5052–0 alu-minum tubing with a wall thick-ness of 0.022”)	135 – 150	60 – 600
5/16”	70 – 90 (30 – 90 for 5052–0 alu-minum tubing with a wall thick-ness of 0.022”)	170 – 200	—
3/8”	110 – 130 (35 – 130 for 5052–0 alu-minum tubing with a wall thickness of 0.022”)	270 – 300	75 – 700
1/2”	230 – 260	450 – 500	100 – 900
5/8”	330 – 360	650 – 700	—
3/4”	460 – 500	900 – 1000	300 – 1600
1”	500 – 700	1200 – 1400	400 – 2200
1 1/4”	800 – 900	1520 – 1680	500 – 2500
1 1/2”	800 – 900	1900 – 2100	—
2”	1800 – 2000	2660 – 2940	—

Note 1 – If the hose fitting material and the tube material are different, use the lower torque value.

Note 2 – For the purpose of this table, size may be considered the tube size, nut size or fitting size, as applicable.

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